

Shuo Guo

School of Data Science, The Chinese University of Hong Kong, Shenzhen
shuoguo@link.cuhk.edu.cn

Education

The Chinese University of Hong Kong, Shenzhen (CUHK-Shenzhen) Shenzhen, China
B.Sc. in Statistics, School of Data Science Sept 2025 – May 2029 (expected)

Experience

Research Assistant May 2026 – Present
Working with PhD student in the research group of Prof. Dr. Ruoyu Sun, CUHK-Shenzhen
Mainly focusing on optimization for Deep Learning.

Research Interests

- Optimization theory
- Optimization for machine learning
- Machine learning theory
- Machine learning from a mathematical perspective

Selected Independent Study / Reading

- **Yurii Nesterov**, *Lectures on Convex Optimization*
Currently reading; studied topics including convexity, gradient descent, and strongly convex complexity bounds.
- **Avrim Blum, John Hopcroft, and Ravindran Kannan**, *Foundations of Data Science*
Read selected chapters related to SVD, random graphs and machine learning.
- **Ruoyu Sun**, *Lectures on Deep Learning Optimization*
- **B. Oksendal**, *Stochastic Differential Equations: An Introduction with Applications*
- **Jean-François Le Gall**, *Measure Theory, Probability, and Stochastic Processes*

Relevant Coursework

- Probability and Statistics
- Linear Algebra
- Calculus
- Computer Science
- Stochastic Analysis

Academic Activities

- Participated in seminars, group meetings, and independent study sessions related to optimization, machine learning, and statistics.

Technical Skills

- Python
- Basic programming for mathematical and statistical computing